



Lab 06: Departmental Subnetting

IT · Finance · HR PCs · Switch3 (2960-24TT) · Router0 (2911)

Lab #06

Topic: Subnetting & VLANs

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Create three VLANs on Switch3 (2960-24TT) for IT, Finance and HR departments.
- Assign each department PC to its VLAN via access ports Fa0/1, Fa0/2 and Fa0/3.
- Configure Fa0/4 on Switch3 as a trunk uplink to Router0 (2911) Gig0/0.
- Set up Router-on-a-Stick sub-interfaces on Router0 for inter-department routing.
- Verify IT, Finance and HR can all reach each other through Router0.

2. Network Topology

Switch3 (2960-24TT) is the central device. IT PC on Fa0/1, Finance PC on Fa0/2, HR PC on Fa0/3. The trunk uplink Fa0/4 connects to Router0 (2911) Gig0/0. Router0 uses sub-interfaces Gig0/0.10, .20 and .30 as the gateway for each department VLAN.

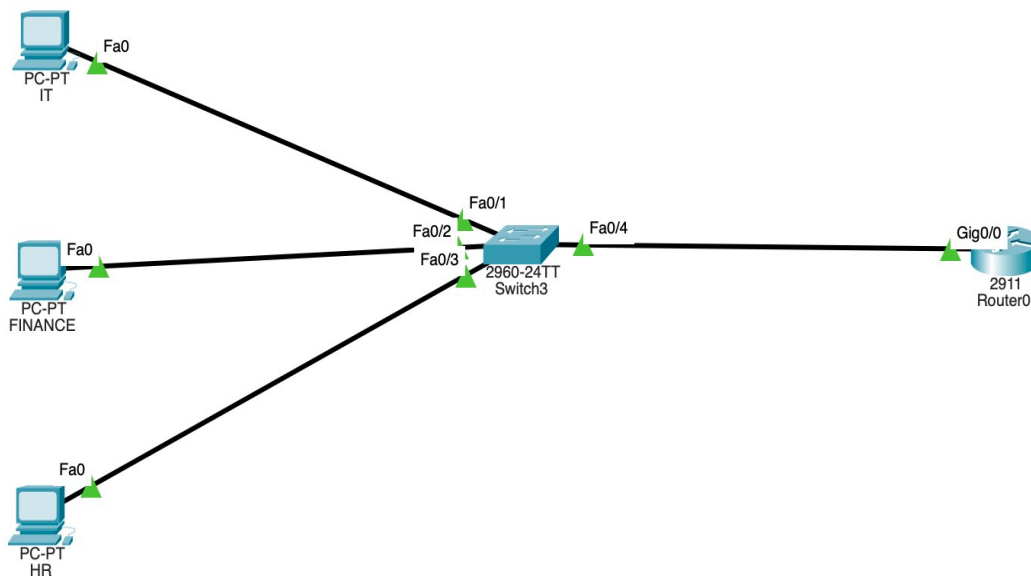


Figure 1 – IT, Finance, HR PCs on Switch3 connected to Router0 via trunk

3. IP Addressing Table

Department	Device	VLAN	Switch Port	IP Address	Subnet Mask	Gateway
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IT	IT	10	Fa0/1	192.168.10.10	255.255.255.0	192.168.10.1
Finance	FINANCE	20	Fa0/2	192.168.20.10	255.255.255.0	192.168.20.1
HR	HR	30	Fa0/3	192.168.30.10	255.255.255.0	192.168.30.1
—	Router0 .10	—	Gig0/0.10	192.168.10.1	255.255.255.0	—
—	Router0 .20	—	Gig0/0.20	192.168.20.1	255.255.255.0	—
—	Router0 .30	—	Gig0/0.30	192.168.30.1	255.255.255.0	—

4. Step-by-Step Configuration

4.1 Switch3 – VLANs, Access Ports & Trunk

```
Switch3>enable
Switch3#configure terminal
Switch3(config)#vlan 10
Switch3(config-vlan)#name IT
Switch3(config-vlan)#exit
Switch3(config)#vlan 20
Switch3(config-vlan)#name FINANCE
Switch3(config-vlan)#exit
Switch3(config)#vlan 30
Switch3(config-vlan)#name HR
Switch3(config-vlan)#exit
Switch3(config)#interface Fa0/1
Switch3(config-if)#switchport mode access
Switch3(config-if)#switchport access vlan 10
Switch3(config-if)#exit
Switch3(config)#interface Fa0/2
Switch3(config-if)#switchport mode access
Switch3(config-if)#switchport access vlan 20
Switch3(config-if)#exit
Switch3(config)#interface Fa0/3
Switch3(config-if)#switchport mode access
Switch3(config-if)#switchport access vlan 30
Switch3(config-if)#exit
Switch3(config)#interface Fa0/4
Switch3(config-if)#switchport mode trunk
Switch3(config-if)#exit
Switch3#write memory
```



4.2 Router0 – Sub-interfaces (Router-on-a-Stick)

```
Router0(config)#interface GigabitEthernet0/0
Router0(config-if)#no shutdown
Router0(config-if)#exit

Router0(config)#interface GigabitEthernet0/0.10
Router0(config-subif)#encapsulation dot1Q 10
Router0(config-subif)#ip address 192.168.10.1 255.255.255.0
Router0(config-subif)#exit

Router0(config)#interface GigabitEthernet0/0.20
Router0(config-subif)#encapsulation dot1Q 20
Router0(config-subif)#ip address 192.168.20.1 255.255.255.0
Router0(config-subif)#exit

Router0(config)#interface GigabitEthernet0/0.30
Router0(config-subif)#encapsulation dot1Q 30
Router0(config-subif)#ip address 192.168.30.1 255.255.255.0
Router0(config-subif)#exit
Router0#write memory
```

5. Verification

```
IT>ping 192.168.20.10      ! IT to Finance – expect Reply
IT>ping 192.168.30.10      ! IT to HR – expect Reply
Finance>ping 192.168.30.10 ! Finance to HR – expect Reply

Switch3#show vlan brief      ! Ports Fa0/1-3 in correct VLANs
Switch3#show interfaces trunk ! Fa0/4 must show as trunk
```

6. Key Takeaways

- VLANs isolate department traffic at Layer 2 — improving both security and performance.
- Router-on-a-Stick routes all VLANs over ONE physical trunk link using 802.1Q sub-interfaces.
- The trunk port on the switch must be configured **before** the router sub-interfaces will work.
- Each department PC's default gateway is its corresponding router sub-interface IP.
- In production, a Layer 3 switch handles inter-VLAN routing in hardware — much faster.